MIL-A-21713(NOrd)

1 December 1958
Issued in lieu of
NAVORD OS 7116

MILITARY SPECIFICATION

AIR DRYER MARK 12 MOD 0

1. SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of an Air Dryer Mark 12 Mod 0, Bureau of Ordnance LD No. 296157. The Air Dryer is a wafer like container inclosing a quantity of silica gel visible through a transparent window provided in the container. The Air Dryer Mark 12 Mod 0 is of one type and one class.

2. APPLICABLE DOCUMENTS

2.1 The following standards, drawings, and publications or such portions thereof as designated herein of the issue in effect on the date of invitation for bids form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-C-6054	Container; Steel, Shipping
MIL-P-3803	Plastic, Polyethylene, Molded and Extruded Shapes, Sheet and Tubing

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-354	Temperature and Humidity Test for Use in Production of Fuzes

FSC 6850

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DRAWINGS

NAVY DEPARTMENT

BUORD LD 296157	Air Dryer Mark 12 Mod 0
BUORD Dwg 1476194	Marking Instructions for Shipping and Storage Containers (Inner and Outer)

And all drawings, standards and specifications listed thereon. In the following specification, all referenced drawing numbers are listed on BUORD LD No. 296157 unless otherwise designated.

AIR FORCE

AN - 8024	Navy Aeronautical Standard Drawings

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for the Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28 and 1950 supplement	Screw thread standards for the Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

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3.1 Description. - The Air Dryer Mark 12 Mod 0 is a wafer like container comprised of two relatively flat cup shaped chipboard half sections with their concave surfaces telescoped and cemented together to form a receptacle for housing a quantity of silica gel. A transparent window closing a centrally formed opening in one of the half sections permit visual inspection of the container contents. The chipboard from which the container is formed is sufficiently porous as to permit communication of moisture to and from the silica gel.

3.2 General requirements

3.2.1 Pre-production samples. - Unless otherwise specified in the contract or order pre-production samples of the Air Dryer Mark 12 Mod 0 shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein and are for the purpose of determining that the production item meets the requirements of this specification and the design. Provision of samples shall be in accordance with 4.1.2 and 4.1.4.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order all materials and purchased parts entering into the final assembly of the air dryer shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUORD LD No. 296157.

3.2.3 Finishes, coatings, construction and dimensions. - Finishes, coatings, construction and dimensions for the air dryer shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUORD LD No. 296157.

3.2.4 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed in BUORD LD No. 296157 and MIL-STD-129.

3.2.5 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings will be made without prior revisions and correction of the applicable drawings.

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3.2.6 Gages. - The Contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy Final Inspection Gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy Final Inspection Gages.

3.2.7 Requests for deviation. - Requests for deviation from this specification, applicable drawings, specification, publications, materials and processes specified herein, shall be as set forth in OP 400.

3.2.8 Conflicting requirements. - Conflicting requirements arising between this specification or any specification, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.3 Definitions. - The following terms will be used throughout this specification of the Air Dryer Mark 12 Mod 0 and are here defined for clarity.

3.3.1 Air dryer. - Reference to the "Air Dryer Mark 12 Mod 0" or "air dryer" shall mean the Air Dryer Mark 12 Mod 0 completely assembled as shown on BUORD Dwg. 1420159.

3.4 Performance requirements and product characteristics. - The Air Dryer Mark 12 Mod 0 shall meet the following performance requirements and product characteristics:

3.4.1 Window securement. - After cementing, the mica window shall be securely held at all points about its edge to the surface of the lid.

3.4.2 Cementing. - The cementing together of the box, lid and window shall be such as to prevent dusting of the silica gel from the container when tested in accordance with paragraph 4.5.4.

3.4.3 Box and lid porosity. - The chipboard box and lid of the Air Dryer Mark 12 Mod 0 shall be sufficiently porous to acceptably meet

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3.5 Environmental requirements

3.5.1 Shock

3.5.1.1 Shock, free fall drop. - The Air Dryer Mark 12 Mod 0 packaged and packed as called for in section 5, shall meet acceptably all performance requirements and product characteristics of this specification after being subjected to the free fall drop test of 4.6.1.1.1.

3.5.1.2 Shock, ready-non-operating. - The Air Dryer Mark 12 Mod 0 shall meet acceptably all performance requirements and product characteristics requirements of this specification after being subjected to the shock test specified in 4.6.1.1.2.

3.5.2 Vibration

3.5.2.1 Transportation, storage and handling. - The Air Dryer Mark 12 Mod 0 shall meet acceptably all performance and product characteristics requirements of this specification after being subjected to the vibration, transportation, storage and handling test of 4.6.1.2.1.

3.5.2.2 Ready-non-operating. - The Air Dryer Mark 12 Mod 0 shall meet acceptably all performance and product characteristics requirements of this specification after being subjected to the vibration, ready-non-operating test of 4.6.1.2.2.

3.5.3 Temperature

3.5.3.1 Operating temperature. - The Air Dryer Mark 12 Mod 0 shall operate in a manner to meet acceptably all performance requirements and product characteristics of the specification at any temperature between ± 28°F and 90°F when tested in accordance with 4.6.1.3.1.

3.5.3.2 Storage temperature. - The Air Dryer Mark 12 Mod 0, packaged as called for in 5, shall operate in a manner to meet acceptably all performance requirements and product characteristics of this specification after being subjected to the 14 day temperature cycling test defined in 4.6.1.3.2.

3.6 Clean-up. - Prior to assembly both box sections and window disk shall be free of frayed, split, or ragged edges. All components shall be kept clean and free of dirt, dust, paint, oil, grease, wax or the like. The silica gel shall be clean and of the proper size and shade.

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3.7 Workmanship. - The air dryer shall be constructed and finished in a manner to assure compliance with all requirements of this specification. During assembly the glue shall be confined to the surface being cemented. Particular attention shall be paid to neatness and thoroughness of workmanship.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that components and assemblies are in accordance with the requirements of the contract, pertinent drawings and specifications. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection shall be in accordance with the provisions set forth in MIL-STD-105.

4.1.2 Pre-production sample. - Unless otherwise specified for each contract or order, a representative sample of twenty-four (24) Air Dryers Mark 12 Mod 0 shall be manufactured using the methods and procedures proposed for the production lot. These dryers shall be submitted to the Government inspector for pre-production tests at an activity designated in the contract or by the Bureau of Ordnance, for purposes of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of tests on pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Lot size. - As applied to Government inspection of units of product the term "lot" shall mean "Inspection lot", i.e. a collection of units of product submitted by a supplier for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" will be as determined by the Government inspector and may differ from the quantity designated in the contract or order in a lot for production, shipment or other purpose.

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4.1.4 Periodic production samples. - From each succeeding lot or grand lot of three hundred (300) or less, a random sample of 8 air dryers will be selected by the Government inspector for submission to a Bureau of Ordnance designated activity to determine compliance of the samples with the requirements of the contract, specification and drawings. An air dryer shall be considered satisfactory if it meets all specifications and pertinent drawing requirements, and withstands the tests as specified in 4.5 and 4.6 inclusive.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of the Air Dryer Mark 12 Mod 0 shall be classified as follows:

4.3.1 Acceptance tests. - These tests are to be accomplished on the air dryers being submitted for acceptance under contract. Inspection and test procedures shall be submitted for Government approval prior to commencement of production. Acceptance tests shall either be performed by the manufacturer and witnessed by the Government inspector, or shall be conducted by the Government inspector. These tests are defined in 4.5.

4.3.2 Pre-production and periodic production tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.4 which are representative of the production of the air dryer after the award of the contract to determine that the lot and production meet the requirements of this specification. These are detailed in 4.6. Acceptance shall be based on no defects in a sample. Failure of a sample to comply with these requirement will result in the rejection of the lot or the cessation of production as determined by the procuring activity.

4.4 Test equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

(a) Means for vibrating the air dryer through a frequency range of from 700 to 3000 cycles per minute.

(b) A black highly polished square plate of approximately 12 x 12 inches.

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4.5 Acceptance tests

4.5.1 The Contractor or Government inspector shall conduct acceptance tests as specified in 4.3.1 to assure that the Air Dryer Mark 12 Mod 0 is in compliance with the requirements of 3.4.1, 3.4.2 and 3.4.3.

4.5.2 Test conditions. - Unless otherwise specified the Air Dryer Mark 12 Mod 0 shall be subjected to acceptance tests under the following conditions:

4.5.2.1 Temperature:	Room ambient 65 degrees F to 95 degrees F
4.5.2.2 Altitude:	Normal ground
4.5.2.3 Vibration:	None
4.5.2.4 Humidity:	Room ambient to 95 percent relative maximum

4.5.3 Test and inspection equipment and facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.5.4 Cementing test. - Prior to this test the assembled Air Dryer Mark 12 Mod 0 shall be vibrated through a range of frequencies from 700 to 3000 cycles per minute in 100 cycles per minute increments. The duration of the vibration of each increment shall be one hour. The amplitude of the vibrations shall be in accordance with the following schedule.

FREQUENCY (cycles per minute)	AMPLITUDE (inches)
700 to 1500	0.030 ± 0.002
1600 to 2000	0.020 ± 0.002
2100 to 2500	0.013 ± 0.001
2600 to 3000	0.009 ± 0.001

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After completing the vibration schedule outlined above, the air dryer shall then be held not more than 6 inches above a clean highly polished black surfaced plate and shaken for a minimum of 15 seconds in a window down position, then 15 seconds in a window up position. The appearance of silica gel dust leaking from the assembly as apparent from a careful examination of the plate, shall be cause for regarding the air dryer defective.

4.5.5 Box and lid porosity test. - Expose the assembled air dryer to the atmosphere until such time as the silica gel becomes pink, indicating a relative humidity within the container of 40% or greater, as viewed through the mica window in the lid. Now place the assembly within a ventilated oven and maintain at 170°F for 24 hours. Any assembly wherein the silica gel fails to turn blue to a valve indicating a relative humidity within the container of 10% or less shall be cause for regarding the air dryer defective. Sampling and acceptance testing for the above characteristic shall be in accordance with MIL-STD-105, Table VI-C AQL 4.0% for any inspection lot size.

4.6 Pre-production and periodic tests

4.6.1 Environmental

4.6.1.1 Shock

4.6.1.1.1 Shock, free fall drop test. - The Air Dryer Mark 12 Mod 0 packaged and packed as designated in 5, shall be subjected to a free fall drop shock from a height of 30 inches to a concrete or hard steel surface. This test shall be applied to each corner of the packaged unit. Upon completion of the test the air dryer shall be capable of meeting acceptably all performance and product characteristics of this specification.

4.6.1.1.2 Shock, ready-non-operating. - The Air Dryer Mark 12 Mod 0 shall be subjected to a continuous two-phase shock applied in any direction with respect to the mechanism.

A. The first (impact) phase shall consist of a change of velocity of 5.5 to 7.0 feet per second taking place in from 0.2 to 0.4 millisecond.

B. During the second (drag) phase, the dryer shall be accelerated for a minimum time of 15 milliseconds in such a manner that during this time the minimum average acceleration is 100 gravity units. The peak acceleration shall be between 125 and 150 gravity units. Upon completion of the above test the air dryer shall be capable of meeting acceptably all performance and other requirements of this specification.

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4.6.1.2 Vibration

4.6.1.2.1 Vibration, transportation, storage and handling. - The Air Dryer Mark 12 Mod 0, packaged and packed in shipping containers as indicated in section 5, shall be mounted on a vibration platform equipped with barriers so as to allow free movement of the container relative to the platform but prevent the devices from falling off the platform. The motion of the platform shall be such that any point on the platform moves in a circular path one inch in diameter with the plane of the circle perpendicular to the platform. The frequency shall be within the range of 240 to 300 cycles per minute. The package of units shall be vibrated for a period of one half hour on each face of the container. After completing the above vibration test, the air dryer shall be capable of meeting acceptably all performance and product characteristics requirements of this specification.

4.6.1.2.2. Vibration, ready-non-operating. - Mount the air dryer rigidly to the vibration table by means simulating the actual method of mounting used in service. The vertical displacement component of the vibration shall be simple harmonic motion. The samples shall be mounted on the vibration table so that equal quantities of devices experience vibration perpendicular to the plane of mounting and vibration parallel to the plane of mounting. The units shall be vibrated through a range of frequencies from 700 to 3000 cycles per minute in 100 cycles per minute increments. The amplitude and duration of the vibration shall be in accordance with the schedule shown in 4.5.4.

4.6.1.2.3 After completing the vibration test of 4.6.1.2.1 and 4.6.1.2.2, the air dryer shall be tested for leakage as called for in 4.5.4. An air dryer wherein silica gel is caused to leak from the container, shall be regarded defective.

4.6.1.3 Temperature

4.6.1.3.1 Operating temperature test. - The Air Dryer Mark 12 Mod 0 shall be performance checked at + 28°F and at 90°F. Any air dryer failing to meet any requirement of this specification at either temperature shall be regarded defective.

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4.6.1.3.2 Storage temperature test. - The air dryer, packaged and packed as designated in 5, shall be subjected to the 14 day temperature cycling test defined in MIL-STD-354. After completing the said test and at ambient temperature, the air dryer shall be tested for operational performance. An air dryer failing thereafter to meet any requirement of this specification shall be regarded defective.

4.7 Failure of the Air Dryer Mark 12 Mod 0 to meet any of the requirements and test of this specification, shall be considered cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level A

5.1.1.1 Twenty - five (25) Air Dryers Mark 12 Mod 0 shall be packaged and heat-sealed in a polyethylene film bag, (4 mil thickness) conforming to the requirements of Military Specification MIL-P-3803. Moisture content of the silica gel shall not exceed 10% at the time of packaging.

5.1.2 Level C (Not applicable, see paragraph 6.1)

5.2 Packing

5.2.1 Level A

5.2.1.1 Eight (8) unit packages, containing a total of two-hundred (200) Air Dryers Mark 12 Mod 0 shall be packed in a Class I reusable gasket-sealed steel shipping container conforming to the requirements of Military Specification MIL-C-6054. For further details on container, see Air Force-Navy Aeronautical Standard Drawing No. AN-8024.

5.2.1.2 Air Dryers Mark 12 Mod 0 shall be stacked in groups of fifty (50), four (4) stacks to the container. A suitable filler material of a non-hygroscopic type shall be used within the shipping container to provide a tight pack.

5.3 Marking

5.3.1 In addition to any special marking required by the contract or order, all marking shall be in accordance with Military Standard MIL-STD-129.

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5.3.1.1 For detailed marking requirements, see Bureau of Ordnance Drawing No. 1476194.

6. NOTES

6.1 Since end use of air dryers are to absorb moisture, maximum packaging and packing protection is required during shipping and storage. Procurement documents shall specify level A packaging and packing.

Notice. When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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or between the tubes. Screen tubes when used shall be considered as the

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8179

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